

Relative exposure to beta-carbolines norharman and harman from foods and tobacco smoke.



Norharman and harman are two heterocyclic beta-carboline (9H-pyrido[3,4-b]indole) alkaloids with biological and potential toxicological activity that appear in foodstuffs and environmental sources. To assess the occurrence and distribution of these compounds and to estimate the exposure levels based on the detected amounts, numerous samples of foodstuffs and cigarette smoke were analysed by solid-phase extraction and high-performance liquid chromatography-fluorescence. The levels found of beta-carbolines were highly variable. Low processed foodstuffs (i.e. milk, yoghurt, uncooked meats and fish) did not contain norharman and harman above the detection limit. Others, however, contained relatively high concentrations (at the tens of ng g⁻¹ or microg l⁻¹ level) depending on the processing conditions as, for example, 'well-done' cooked meat and fish. The highest amounts of norharman and harman were found in brewed coffee (29-207 microg l⁻¹), sauces (soy sauce and Tabasco, among others; 4-252 microg l⁻¹), 'well done' cooked meat and fish (57-160 ng g⁻¹), toasted bread (42-160 ng g⁻¹), and fermented alcoholic beverages (n.d.-41 mug l⁻¹). beta-Carbolines also occurred in a high amount in the mainstream of cigarette smoke (207-2780 ng/cigarette), which is an important contributor to daily exposure to these compounds. Based on these results, it is concluded that the daily exposure to beta-carbolines in humans might be from tens to hundreds of micrograms, with cigarette smoke, coffee, certain seasonings, cooked foods and alcoholic beverages, in this order, being the major contributors. Many other foodstuffs might also contribute with minor amounts of norharman and harman. Foods and tobacco smoke might be potential contributors to the reported endogenous presence of beta-carbolines in humans.